STACKS WITH QUASI-AFFINE DIAGONAL SATISFY FPQC DESCENT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4320-4329 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 4331-4457, the complete author proof (125 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 4 of queue_stacks-stacks-more-morphisms.json (pb=4561, stmt 4320-4329, proof 4331-4457).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Editorial note. Statement carries a reference/slogan/history environment (patchenv -> patch-sd-envs.tex).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original proposition 0247: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4457–4457. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0APK: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4457–4457. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0APL: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4457–4457. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 02ZC: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4457–4457. Cited by the retained statement or

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Proposition 0.1. *Proposition 3.3.6 of "Intro to Algebraic Stacks" by Anatoly Preygel. Let $\mathcal{X}$ be an algebraic stack with quasi-affine[1] diagonal. Then $\mathcal{X}$ satisfies descent for fpqc coverings.*

Proof. Our conventions are that $\mathcal{X}$ is a stack in groupoids $p : \mathcal{X} \to (Sch/S)_{fppf}$ over the category of schemes over a base scheme S endowed with the fppf topology. The statement means the following: given an fpqc covering $\mathcal{U} = \{U_i \to U\}_{i \in I}$ of schemes over S the functor

$$\mathcal{X}_U \longrightarrow DD(\mathcal{U})$$

is an equivalence. Here on the left we have the category of objects of $\mathcal{X}$ over U and on the right we have the category of descent data in $\mathcal{X}$ relative to $\mathcal{U}$. See discussion in Stacks, Section 02ZC.

Fully faithfulness. Suppose we have two objects x, y of $\mathcal{X}$ over U. Then $I = Isom(x, y)$ is an algebraic space over U. Hence a collection of sections of I over U_i whose restrictions to $U_i \times_U U_j$ agree, come from a unique section over U by the analogue of the proposition for algebraic spaces, see Properties of Spaces, Proposition 0APL. Thus our functor is fully faithful.

Essential surjectivity. Here we are given objects x_i over U_i and isomorphisms $\varphi_{ij} : \mathrm{pr}_0^* x_i \to \mathrm{pr}_1^* x_j$ over $U_i \times_U U_j$ satisfying the cocyle condition over $U_i \times_U U_j \times_U U_k$.

Let W be an affine scheme and let $W \to \mathcal{X}$ be a morphism. For each i we can form

$$W_i = U_i \times_{x_i, \mathcal{X}} W$$

The projection $W_i \to U_i$ is quasi-affine as the diagonal of $\mathcal{X}$ is quasi-affine. For each pair $i, j \in I$ the isomorphism φ_{ij} induces an isomorphism

$$W_i \times_U U_j = (U_i \times_U U_j) \times_{x_i \circ \mathrm{pr}_0, \mathcal{X}} W \to (U_i \times_U U_j) \times_{x_j \circ \mathrm{pr}_1, \mathcal{X}} W = U_i \times_U W_j$$

Moreover, these isomorphisms satisfy the cocycle condition over $U_i \times_U U_j \times_U U_k$. In other words, these isomorphisms define a descent datum on the schemes W_i/U_i relative to $\mathcal{U}$. By Descent, Lemma 0247 we see that this descent datum is effective[2]. We conclude that there exists a quasi-affine morphism $W' \to U$ and a commutative diagram

$$\begin{array}{ccccc} W' & \longleftarrow & W_i & \longrightarrow & W \\ \downarrow & & \downarrow & & \downarrow \\ U & \longleftarrow & U_i & \xrightarrow{x_i} & \mathcal{X} \end{array}$$

whose squares are cartesian. Since $\{W_i \to W'\}_{i \in I}$ is the base change of $\mathcal{U}$ by $W' \to U$ we conclude that it is an fpqc covering. Since W satisfies the sheaf condition for fpqc coverings, we obtain a unique morphism $W' \to W$ such that $W_i \to W' \to W$ is the given morphism $W_i \to W$. In other words, we have the commutative diagrams

$$\begin{array}{ccccc} W_i & \longrightarrow & W' & \longrightarrow & W \\ \downarrow & & \downarrow & & \downarrow \\ U_i & \longrightarrow & U & & \mathcal{X} \\ & \searrow^{x_i} & & \nearrow & \end{array}$$

compatible with the isomorphisms φ_{ij} and whose square and rectangle are cartesian.

[1] It suffices to assume ind-quasi-affine.

[2] Or use More on Groupoids, Lemma 0APK in the case of ind-quasi-affine diagonal.

Choose a collection of affine schemes W_α, $\alpha \in A$ and smooth morphisms $W_\alpha \to \mathcal{X}$ such that $\coprod W_\alpha \to \mathcal{X}$ is surjective. By the procedure of the preceding paragraph we produce a diagram

$$\begin{array}{ccccc} W_{\alpha,i} & \longrightarrow & W'_\alpha & \longrightarrow & W_\alpha \\ \downarrow & & \downarrow & & \downarrow \\ U_i & \longrightarrow & U & & \mathcal{X} \\ & & & \overset{x_i}{\searrow} & \end{array}$$

for each α. Then the morphisms $W'_\alpha \to U$ are smooth and jointly surjective.

Denote x_α the object of $\mathcal{X}$ over W'_α corresponding to $W'_\alpha \to W_\alpha \to \mathcal{X}$. Since $\mathcal{X}$ is an fppf stack and since $\{W'_\alpha \to U\}$ is an fppf covering, it suffices to show that there are isomorphisms $\mathrm{pr}_0^* x_\alpha \to \mathrm{pr}_1^* x_\beta$ over $W'_\alpha \times_U W'_\beta$ satisfying the cocycle condition. However, after pulling back to $W_{\alpha,i}$ we do have such isomorphisms over $W_{\alpha,i} \times_{U_i} W_{\beta,i} = U_i \times_U (W'_\alpha \times_U W'_\beta)$ since the pullback of x_α to $W_{\alpha,i}$ is isomorphic to the pullback of x_i to $W_{\alpha,i}$. Since $\{U_i \times_U (W'_\alpha \times_U W'_\beta) \to W'_\alpha \times_U W'_\beta\}_{i \in I}$ is an fpqc covering and by the aforementioned compatibility of the diagrams above with φ_{ij} these isomorphisms descend to $W'_\alpha \times_U W'_\beta$ and the proof is complete. $\square$